Robert Dougherty-Bliss

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Research interests

Experimental mathematics; symbolic-numeric computation; difference equations; combinatorics; number theory

Academic positions

Dartmouth College
Instructor of Applied and Computational Mathematics *2024–*
Mentor: Peter Winkler

Johannes Kepler University Linz
Visitor *August 2023*
Host: Manuel Kauers

Education

Rutgers University — New Brunswick
Ph.D. Mathematics *2019–2024*
Advisor: Doron Zeilberger

Oglethorpe University
B.S. Mathematics; Computer Science *2014–2019*
Summa cum laude, with honors

Publications

[1] Pablo Blanco, Robert Dougherty-Bliss, Natalya Ter-Saakov, and Doron Zeilberger. “In How Many Ways can a Rectangle be Rectangled?” In: *arXiv preprint arXiv:2606.05439* (2026).

[2] Robert Dougherty-Bliss and Sergi Elizalde. “Necklaces, subset sums, and cyclic permutations”. In: *arXiv preprint arXiv:2603.15830* (2026).

[3] Robert Dougherty-Bliss, Alejandro B Galván, Michaela A Polley, and David Shuster. “Enumerating separable derangements”. In: *arXiv preprint arXiv:2608.27583* (2026).

[4] Robert Dougherty-Bliss, Natalya Ter-Saakov, Mits Kobayashi, and Eugene Zima. “Dyadically Resolving Trinomials for Fast Modular Arithmetic”. In: *Math. Comput. Sci.* 20.1 (2026), Paper No. 18.

[5] Robert Dougherty-Bliss, Charles Kenney, and Doron Zeilberger. “Creating Decidable Diophantine Equations”. In: *The American Mathematical Monthly* 132.10 (2025), pp. 947–959.

[6] Robert Dougherty-Bliss, Mits Kobayashi, Natalya Ter-Saakov, and Eugene Zima. “On relative primality and other properties of trinomials”. In: *6th International Conference «Computer Algebra», Moscow*. 2025, pp. 47–50.

[7] Robert Dougherty-Bliss, Christoph Koutschan, Natalya Ter-Saakov, and Doron Zeilberger. “The (Symbolic and Numeric) Computational Challenges of Counting 0-1 Balanced Matrices”. In: *Enumer. Comb. Appl.* 5.2 (2025), Paper No. S2R14, 8.

[8] Robert Dougherty-Bliss and Natalya Ter-Saakov. “The Comma Sequence is Finite in Other Bases”. In: *J. Integer Seq.* 28.5 (2025), Art. 25.5.5, 18.

[9] Robert Dougherty-Bliss, Natalya Ter-Saakov, and Doron Zeilberger. "Cutting 4 by n grids into two congruent pieces". 2025.

[10] Robert Dougherty-Bliss and Manuel Kauers. "Hardinian Arrays". In: *Electronic Journal of Combinatorics* 31.2 (2024), P2.9.

[11] Robert Dougherty-Bliss and Doron Zeilberger. "Automatic conjecturing and proving of exact values of some infinite families of infinite continued fractions". In: *The Ramanujan Journal* 61.1 (2023), pp. 31–47.

[12] Robert Dougherty-Bliss and Doron Zeilberger. "Exploring general Apéry limits via the Zudilin–Straub t-transform". In: *Journal of Difference Equations and Applications* 29.1 (2023), pp. 32–42.

[13] Robert Dougherty-Bliss and Doron Zeilberger. "Lots and Lots of Perrin-Type Primality Tests and Their Pseudo-Primes". In: *Integers* 23 (2023), A95.

[14] Robert Dougherty-Bliss. "Integral Recurrences from A to Z". In: *The American Mathematical Monthly* 129.9 (2022), pp. 805–815.

[15] Robert Dougherty-Bliss. "The Meta-C-Finite Ansatz". In: *Fibonacci Quart* 60.5 (2022), pp. 143–150.

[16] Robert Dougherty-Bliss, Christoph Koutschan, and Doron Zeilberger. "Tweaking the Beukers integrals in search of more miraculous irrationality proofs à la Apéry". In: *The Ramanujan Journal* 58.3 (2022), pp. 973–994.

[17] AJ Bu and Robert Dougherty-Bliss. "Enumerating restricted Dyck paths with context-free grammars". In: *Integers* 21 (2021), A69.

[18] Robert Dougherty-Bliss and Doron Zeilberger. "Experimenting with Apéry Limits and WZ pairs". In: *Maple Transactions* 1.2 (2021), Article 14359.

[19] K. Abernathy, Z. Abernathy, R. Dougherty-Bliss, C. Mayer, and H. Whiteside. "Global dynamics of a cancer stem cell treatment model". In: *International Journal of Dynamical Systems and Differential Equations* 9.2 (2019), pp. 176–186.

Research talks and presentations

[1] *Enumerating Balanced Matrices*. International Congress on Mathematical Software. 2026.

[2] *Necklaces, subset sums, and permutations*. Dartmouth Combinatorics Seminar. 2026.

[3] *p-adically resolving trinomials*. 6th International Conference "Computer Algebra", Moscow (online). 2025.

[4] *Enumerating Balanced Matrices*. MAA MathFest, Sacramento. 2025.

[5] *Modular arithmetic with trinomial moduli*. Rutgers Experimental Mathematics Seminar. 2025.

[6] *Modular arithmetic with trinomial moduli*. Waterloo Symbolic Computation Seminar. 2025.

[7] *On relative primality and other properties of trinomials*. *Computer Assisted Mathematics*, St. Petersburg (online). 2025.

[8] *Balanced Matrices*. Dartmouth Combinatorics Seminar. 2024.

[9] *Enumerating Balanced Matrices*. Differential Algebra and Related Topics XII (Beijing). 2024.

[10] *Rectangular Hardinian Arrays*. Poster at ISSAC. 2024.

[11] *The Comma Sequence is Finite in Other Bases*. Fall AMS Central Sectional. 2024.

[12] *New primality tests and their (very large) pseudoprimes*. Institute für Algebra Seminar. Linz, Austria, 2023.

[13] *New primality tests and their (very large) pseudoprimes*. Fall AMS Southeastern Sectional. Mobile, Alabama, 2023.

[14] *Gosper's algorithm and Bell numbers*. Fall AMS Southeastern Sectional. Chattanooga, Tennessee, 2022.

[15] *The Meta C-finite ansatz*. 20th International Fibonacci Conference. Sarajevo, Bosna i Hercegovina, 2022.

[16] *The Meta-C-finite ansatz*. Rutgers Experimental Mathematics Seminar. 2022.

[17] *Combinatorial Union Busting with Stopped Strings*. New York Combinatorics Seminar. 2021.

[18] *More irrationally good approximations from Beukers integrals*. Combinatorial and Additive Number Theory. 2021.

[19] *Automagic Inverse Continued Fraction Calculators*. Rutgers Experimental Mathematics Seminar. 2020.

[20] *The Ergonomics of Computer Algebra*. Rutgers Experimental Mathematics Seminar. 2019.

Outreach talks

[1] *Me and my friend Claude: A small case study in AI-assisted mathematical research*. MAA MathFest. 2026.

[2] *Mersenne Numbers, Cardioids, and Graphs*. UVM–Dartmouth Math Day (won best talk award). 2026.

[3] *How to compute with big numbers*. Rutgers Undergraduate Mathematical Association. 2025.

[4] *What to do when you don't know*. Dartmouth Mathematical Society. 2025.

[5] *Identities*. Baruch College REU. 2024.

[6] *Proofs by Example*. DIMACS / Baruch College REU. 2023.

[7] *Proofs by Example*. Rutgers Undergraduate Mathematical Association. 2023.

[8] *When does a sum have an answer?* Rutgers Undergraduate Mathematical Association. 2022.

[9] *Mathematical biology for fun and profit*. Central Washington University Math Club. 2021.

Teaching

Dartmouth College

Linear Algebra, Differential Equations, Multivariable Calculus, Introduction to Calculus, Probability
Reading Course in Mathematics (supervised Elsa Taylor-Lillquist twice, Pavan Kumar)

Rutgers University

As instructor: Number Theory (Summer 2021, 2022)
As teaching assistant: Real Analysis (Spring 2021, Fall 2021, Fall 2023), Abstract Algebra (Fall 2023), Calculus III (Fall 2022), Calculus II (Spring 2022, Spring 2023), Calculus I (Fall 2020)

Professional service

Invited Paper Session: Computers and Combinatorics — **MAA MathFest**
Co-organizer — *2026*

Invited Paper Session: Combinatorics and Probability — **MAA MathFest**
Co-organizer — *2026*

Symbolic Combinatorics Minisymposium — **SIAM AG**
Co-organizer — *2025*

Experimental Mathematics Seminar — **Rutgers University**
Co-organizer — *2020–2024*

American Mathematical Society Graduate Chapter — **Rutgers University**
President — *2020–2024*

Undergraduate Directed Reading Program — **Rutgers University**
Mentor — *2023*

Arkansas Undergraduate Mathematics Competition
Problem writer / grader — *2023*

DIMACS REU
Graduate school panelist — *2022*

MoMath Masters — **National Museum of Mathematics**
Problem writer — *2021*

Graduate Liaison Committee — **Rutgers University**
Member — *2020–2022*

Combinatorics and Algebras from A to Z
Co-organizer — *2021*
Birthday conference in honor of Doron Zeilberger and Amitai Regev.

Reviewer for: Journal of Number Theory, The American Statistician, Discrete Mathematics, Discrete Mathematics Letters, Journal of Computational Algebra, European Journal of Combinatorics

Awards

Rutgers University

Excellence in Leadership and Teaching Award — *2024*
University-wide award for leadership and teaching.

TA Teaching Excellence Award — *2022, 2023*
In recognition of valuable contributions to undergraduate teaching in the math department.

Oglethorpe University

James Edward Oglethorpe Award — *2019*
In recognition of scholarship, leadership, and character. Awarded by university faculty vote.

Sally Hull Weltner Award for Scholarship — *2019*
Awarded to the senior with the highest grade-point average.

Ronald L. Carlisle Prize — *2019*
Given to an outstanding senior in mathematics and computer science.

Past Positions

Data Scientist
Government Accountability Office 🇺🇸 — *2023*

Statistical Consultant
League of Women Voters of New Jersey — *2022*

Summer Fellow and Instructor
Metric Geometry and Gerrymandering Group, Tufts University and MIT — *2018, 2019*

Summer Instructor
Oglethorpe University — *2017*

Summer Fellow
Winthrop University — *2016*